Tutorial

A university wants to predict the final exam score of students based on their academic and behavioural attributes data from 10 students has been collected

Input features:

- Study hours per week
- Attendance percentage
- Assignments Score (out of 100)
- No. of practice Tests Taken

Target Variable:

- Final exam score (out of 100)

Questions:

1. Identify Indep. dependant Variables.
2. Write M.L. regn eqn
3. Explain impact of attendance on exam score
4. State one assumption of L. Regn in this scenario
   Find MSE for 5 iterations.

$\hat{y}_4 = 3.95 + (248.5)\,18 + (1639.25)\,95 + (1.512.85)\,87 +$

$(59.25)\,5 = 293632.75$

$\hat{y}_5 = 3.95 + (248.5)\,8 + (1639.25)\,70 + (1,512.85)\,65 +$

$(59.25)\,1 = 215133.95$

y	$\hat{y}$	error
80	260502.25	− 260422.25
85	275554.65	− 275469.65
72	231450.7	− 231378.07
90	293632.75	− 293542.75
68	215133.95	− 215065.95

$MSE = \frac{1}{5} \Sigma (y - \hat{y})^2$

$= \frac{1}{5} \left((260422.25)^2 + (275469.65)^2 + (231378.07)^2 + (293542.75)^2 + (215065.95)^2 \right)$

$\Rightarrow 6.5932 \times 10^{10}$ $\quad MSE = \frac{1}{n} \Sigma (y - \hat{y})^2$

MSE

$w_2 = 0.02 - (0.1)(-0.4767)(1) = 0.027$

$w_3 = 0.02 - (0.1)(-0.4767)(1) = 0.027$

$w_4 = 0.03 - (0.1)(-0.4767)(1) = 0.012$

1. Independent variables

x_1 = study hours

x_2 = Attendance %

x_3 = Assignment Score

x_4 = practice tests

y = ~~practi~~ final exam score.

2. $y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4$

$y = b_0 + b_1$ (Study hrs) $+ b_2$ (Att %) $+ b_3$ (Ass score) $+$ b_4 (practice test)

3. As attendance increases exam score also ↑esg

∴ There is a +ve rel b/w attendance and exam score (i.e $\beta_2 > 0$)

4. There is a linear Rel^n b/w Input features and final exam score

let us assume weights as 0

i.e $w_1, w_2, w_3, w_4 = 0$

and learning rate = 0.01

$w_2 = 0.04 - (0.1)(0.706)(0.18) = 0.02$

$w_3 = 0.04 - (0.1)(0.706)(0.18) = 0.02$

$w_4 = 0.04 - (0.1)(0.706)(0.09) = 0.03$

Sample ③

$z = -0.0116 + (1)(0.035) + (1)(0.02) + (1)(0.02) + (1)(0.03) = 0.0934$

$$\hat{y} = \frac{1}{1+e^{-z}} = 0.5233$$

loss function

$$J = [(1)\log(0.5233) + (1-1)(\log(1-0.5233)]$$

$$= -[\log(0.5233)] = 0.2812.$$

error $(\hat{y} - y)$

$= 0.5233 - 1 = -0.4767$

<u>gradient descent</u>

$w_0 = -0.0116 - (0.1)(-0.4767)(1) = 0.0354$

$w_1 = 0.035 - (0.1)(-0.4767)(1) = 0.012$

Q) A university wants to predict whether a student will pass or fail an exam using SVM

Input features.

- study hrs per week
- Attendance (%)
- Assignment score
- Internal test marks

Target variable : class (pass = +1 , fail = -1)

Study hrs	Attendance	Assignment	Internal	class
14	88	80	35	+1
6	60	55	22	-1
16	92	85	40	+1
5	58	50	20	-1

Questions:

1. Define the hyperplane eqn for linear SVM.
2. what happens when a data point violates the margin?
3. compute the accuracy of the SVM model.
4. Given predicted class labels, calculate classification accuracy for 4 iterations with weight updation.

$f(x) = -31.15$

$f(x) \leq 0 \quad = -1$

Accuracy

new y	old y
-1	1
-1	-1
-1	1
-1	-1

Accuracy $= 2/4 \times 100 = 50\%$

Margin

$$\Rightarrow \frac{2}{\sqrt{(0.03)^2 + (-0.80)^2 + (-0.28)^2 + (-0.07)^2}}$$

$$\Rightarrow \frac{2}{\sqrt{0.01583}}$$

$$\Rightarrow 5.026776 0614 \approx 85.02$$

$= 30 - 0.001 \left(\frac{15015.3}{5} \right)$

$\theta_0 = 26.98$

$\theta_1 = \theta_1 - \alpha \left(\frac{1}{n} \Sigma (y^n - y) x_1 + 2\lambda \theta_1 \right]$

$\Sigma \text{error}_1 = 2792.9(1500) + 3340.1 (1800) + 2269.6 (1200)$

$+ 3687 (2000) + 3005.7 (1600)$

$\frac{1}{n} \Sigma \text{error } x_1 = \frac{25108170}{5} = 5021634$

$\theta_1 = 2 - 0.001 \ (5021634 + 2(0.8)(2)) +$

$3687(4) + 3005.7(3) = 46703.3$

$\frac{1}{n} \Sigma \text{error } x_2 = \frac{46703.3}{5} = 9340.66$

$\theta_2 = 0.3 - 0.001 \left[9340.66 + 2(0.8)(0.3) \right]$

$\boxed{\theta_2 = -9.04114}$

$\theta_3 = \theta_3 - \alpha \left(\frac{1}{n} \Sigma (y^n - y) x_3 + 2 \times \theta_3 \right)$

$\Sigma (y^n - y) x_3 = 2792.9 (5) + 3340.1 (3) + 2269.6 (10)$

$+ 3687 (2) + 3005.7 (7) = 75,0947.7$

$\frac{1}{n} \Sigma (y^n - y) x_3 = 15,018.94$

$w^T x + b = -1$ -ve hyperplane

$w^T x + b = +1$ +ve hyperplane.

$$\text{Margin} = \frac{2}{\|w\|} \qquad \|w\| = \sqrt{w_1^2 + w_2^2 + \cdots w_n^2}$$

Step-3: Train the SVM using 4 iterations

updation rule:

$\Rightarrow$ If

$y \cdot w^T x + b \geq 1 \rightarrow$ correct

$y \cdot w^T x + b < 1 \longrightarrow$ incorrect

weight updation.

$w = w + \eta y x_i$

Ist iteration:

$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b$

$w = (0, 0, 0, 0) + (0.01)(1)[14, 88, 80, 35]$

$= [0.14, 0.88, 0.8, 0.35]$

2nd iteration:

$= (0.14)(6) + (0.0)(0.88) + (0.8)(55) + (0.35)(22)$

$= 105.34$

$y f(x) \leq 1$

update the weights.

$w_1 = 0.08$ $w_2 = 0.28$ $w_3 = 0.25$ $w_4 = 0.13$

3rd iteration [16, 92, 88, 40]

$f(x) = [0.08][16] + [0.28](92) + (0.25)(85) + (40)(0.13) = 50.73$

$(1)(50.73) \geq 1 \rightarrow$ correct

no updation.

4th iteration: [5, 58, 50, 20]

$f(x) = [0.08][5] + [0.28][58] + [50][0.25] + [0.13][20]$.

$= 31.74$.

$y f(x) = (-1)(31.74) < 0$

updation of value:

$w = [0.08, 0.28, 0.25, 0.13)] + (-1)(0.01)$

$[5, 58, 50, 20]$

$= [0.03, -0.3, -0.25, -0.07]$

$x = [14, 88, 80, 35]$

$f(x) = 0.03[14] + (-0.30)(88) + (-0.25)(80) + (-0.07)(35)$

$f(x) = -48.43$

$f(x) < 0 \rightarrow y = -1$

$\therefore y = -1$

$x = [6, 60, 55, 22]$

$f(x) = 0.03(6) + (-0.30)(60) + (-0.25)(55) + (-0.07)(22)$

$f(x) = -33.11$

$f(x) \leq 0$

$\therefore y = -1$

$x = [16, 92, 85, 40]$

$f(x) = 0.03(16) + (-0.30)(60) + (-0.25)(55) + (-0.07)(40)$

$f(x) = -51.47$

$f(x) \leq 0 = -1$

$x = [5, 58, 50, 20]$

$f(x) = 0.03(5) + (-0.30)(98) + (-0.25)(50) + (-0.07)(40)$